

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281262

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Salmon; Fay T. et al.

Method for Generating Design of Orthodontic Appliance

Abstract

A method for generating a design of an orthodontic appliance includes receiving a digital three-dimensional (3D) representation of at least one tooth. The method further includes accessing a Reduced Order Model (ROM) of a periodontal ligament (PDL) of the at least one tooth in the digital 3D representation. The ROM represents a mechanical response of the PDL of the at least one tooth. The method further includes generating the design of the orthodontic appliance based at least partially on the ROM of the PDL of the at least one tooth.

Inventors: Salmon; Fay T. (Eden Prairie, MN), Geisler; Karl J. L. (Saint Paul, MN), Fan; Jiadi (Woodbury, MN), Shen; Tong (Woodbury, MN), Lai; Ming-Lai (Afton, MN), Liu; Jialin (St Louis Park, MN), Grambow; Colin A. (Maplewood, MN)

Applicant: Solventum Intellectual Properties Company (Maplewood, MN)

Family ID: 1000008671770

Appl. No.: 18/861280

Filed (or PCT Filed): April 18, 2023

PCT No.: PCT/IB2023/053957

Related U.S. Application Data

us-provisional-application US 63338244 20220504

Publication Classification

Int. Cl.: A61C7/00 (20060101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to a method for generating a design of an orthodontic appliance for an orthodontic treatment.

BACKGROUND

[0002] Orthodontic treatments are conducted by dental practitioners for moving one or more teeth of a patient from a malposition to a desired position in a patient. The orthodontic treatments may improve a facial appearance of the patient. In some cases, the orthodontic treatments may also improve function of the one or more teeth by providing improved occlusion during mastication. In some cases, orthodontic appliances, for example, orthodontic brackets and aligners, may be used in the orthodontic treatments. Positions of the one or more teeth during the orthodontic treatment may represent a balancing-act between forces and/or moments delivered by the orthodontic appliances and mechanical responses of the one or more teeth, specifically the mechanical response of a periodontal ligament of each of the one or more teeth anchored into the mandibular/maxillary bone.

SUMMARY

[0003] In an aspect, the present disclosure provides a method for generating a design of an orthodontic appliance. The method includes receiving a digital three-dimensional (3D) representation of at least one tooth. The method further includes accessing a Reduced Order Model (ROM) of a periodontal ligament (PDL) of the at least one tooth in the digital 3D representation. The ROM represents a mechanical response of the PDL of the at least one tooth. The method further includes generating the design of the orthodontic appliance based at least partially on the ROM of the PDL of the at least one tooth.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Exemplary embodiments disclosed herein may be more completely understood in consideration of the following detailed description in connection with the following figures. The figures are not necessarily drawn to scale. Like numbers used in the figures refer to like components. However, it will be understood that the use of a number to refer to a component in a given figure is not intended to limit the component in another figure labeled with the same number.

[0005] FIG. 1 is a schematic view of a digital three-dimensional (3D) representation of a dental arch of a patient undergoing an orthodontic treatment, according to an embodiment of the present disclosure;

[0006] FIG. 2 is a schematic sectional view of at least one tooth and a periodontal ligament (PDL) of the at least one tooth, according to an embodiment of the present disclosure;

[0007] FIG. 3 is a schematic block diagram of a system for generating a design of an orthodontic appliance, according to an embodiment of the present disclosure;

[0008] FIG. 4 is a schematic view of a Reduced Order Model (ROM) of the PDL of the at least one tooth, according to an embodiment of the present disclosure;

[0009] FIG. 5 is a schematic diagram of a physics-based mathematical model (PMM) of the at least one tooth, according to an embodiment of the present disclosure;

[0010] FIG. 6A is a schematic isometric view of the PMM and an orthodontic force parameter

applied at a reference point, according to an embodiment of the present disclosure;

[0011] FIG. 6B is a schematic side view of the PMM upon application of the orthodontic force parameter at the reference point, according to an embodiment of the present disclosure;

[0012] FIG. 6C is a schematic front view of the PMM upon application of the orthodontic force parameter at the reference point, according to an embodiment of the present disclosure;

[0013] FIG. 6D is a schematic top view of the PMM upon application of the orthodontic force parameter at the reference point, according to an embodiment of the present disclosure;

[0014] FIG. 7 is a schematic diagram of an updated PMM of the at least one tooth, according to an embodiment of the present disclosure;

[0015] FIG. 8 is a schematic diagram of an updated PMM of the at least one tooth, according to another embodiment of the present disclosure;

[0016] FIG. 9 is a schematic diagram of a design of the dental arch of the patient undergoing the orthodontic treatment having a target arrangement of the at least one tooth, according to an embodiment of the present disclosure;

[0017] FIG. 10 is a schematic diagram of the design of the orthodontic appliance and the dental arch of the patient, according to an embodiment of the present disclosure;

[0018] FIG. 11 is a schematic diagram of the updated PMM and a virtual orthodontic appliance design, according to an embodiment of the present disclosure;

[0019] FIG. 12 is a schematic diagram of a design of the dental arch of the patient undergoing the orthodontic treatment having a resulting arrangement of the at least one tooth, according to an embodiment of the present disclosure;

[0020] FIG. 13 is schematic view of a PMM, according to another embodiment of the present disclosure;

[0021] FIG. 14 is a schematic view of a two-dimensional (2D) image of the PDL of the at least one tooth, according to another embodiment of the present disclosure;

[0022] FIG. 15 is schematic view of a PMM, according to another embodiment of the present disclosure;

[0023] FIG. 16 is a flowchart of a method for generating the design of the orthodontic appliance, according to an embodiment of the present disclosure; and

[0024] FIGS. 17A-17J depict a schematic view of a method for generating the design of the orthodontic appliance, according to an embodiment of the present disclosure

DETAILED DESCRIPTION

[0025] In the following description, reference is made to the accompanying figures that form a part thereof and in which various embodiments are shown by way of illustration. It is to be understood that other embodiments are contemplated and may be made without departing from the scope or spirit of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense.

[0026] In the following disclosure, the following definitions are adopted.

[0027] As used herein, all numbers should be considered modified by the term “about”. As used herein, “a,” “an,” “the,” “at least one,” and “one or more” are used interchangeably.

[0028] As used herein as a modifier to a property or attribute, the term “generally”, unless otherwise specifically defined, means that the property or attribute would be readily recognizable by a person of ordinary skill but without requiring absolute precision or a perfect match (e.g., within $\pm 20\%$ for quantifiable properties).

[0029] The term “substantially”, unless otherwise specifically defined, means to a high degree of approximation (e.g., within $\pm 10\%$ for quantifiable properties) but again without requiring absolute precision or a perfect match.

[0030] The term “about”, unless otherwise specifically defined, means to a high degree of approximation (e.g., within $\pm 5\%$ for quantifiable properties) but again without requiring absolute precision or a perfect match.

[0031] Terms such as same, equal, uniform, constant, strictly, and the like, are understood to be within the usual tolerances or measuring error applicable to the particular circumstance rather than requiring absolute precision or a perfect match.

[0032] As used herein, the terms “first” and “second” are used as identifiers. Therefore, such terms should not be construed as limiting of this disclosure. The terms “first” and “second” when used in conjunction with a feature or an element can be interchanged throughout the embodiments of this disclosure.

[0033] As used herein, “at least one of A and B” should be understood to mean “only A, only B, or both A and B”.

[0034] As used herein, the term “three-dimensional representation,” refers to any three-dimensional surface map of an object, such as a point cloud of surface data, a set of two-dimensional polygons, or any other data representing all or some of the surface of an object, as might be obtained through the capture and/or processing of three-dimensional scan data, unless a different meaning is explicitly provided or otherwise clear from the context. A “three-dimensional representation” may include volumetric and other representations, unless a different meaning is explicitly provided or otherwise clear from the context.

[0035] As used herein, the term “Reduced order model (ROM)” refers to simplification of a high-fidelity, complex model. The ROM captures the behavior of a source model so that a system's dominant effects can be studied using minimal computational resources.

[0036] As used herein, the term “physics-based mathematical model (PMM)” refers to a model representation based on general laws of physics and mathematical concepts, formulas, or equations.

[0037] As used herein, the term “center of resistance (CoR)” refers to a point in the root region of a tooth at which resistance to movement of a tooth and PDL system can be concentrated for analysis. Conceptually, translational forces applied at the CoR only produce translational displacements of the tooth, and rotational forces (moments) applied at the CoR only produce rotational displacements of the tooth.

[0038] The CoR is typically located in a root region of the tooth and may be inside or outside the root volume. depending on root shape and quantity. In reality, the COR may be different for different forces, and the CoR concept may be more accurately described as a small volume in 3-dimensional space or average CoR point.

[0039] As used herein. the term “finite element analysis (FEA)” may refer to simulation of any given physical phenomenon using a numerical technique called Finite Element Method (FEM). The FEM is a systematic procedure of approximating continuous functions as discrete models.

[0040] As used herein, the term “calibration” may be defined as finding a unique set of model parameters that provide a good description of a system behavior, and can be achieved by confronting model predictions with measurements performed on the system.

[0041] Conventionally, finite-element methods (FEMs) are typically used for simulation of orthodontic biomechanics. Positions of the one or more teeth during the orthodontic treatment may represent a balancing-act between forces and moments delivered by orthodontic appliances and mechanical responses of the one or more teeth, specifically the mechanical response of a periodontal ligament (PDL) of each of the one or more teeth anchored into the mandibular/maxillary bone. As a result, finite-element models generated by the FEMs include a detailed representation of the PDLs.

[0042] However, due to concerns about patient and staff radiation exposure, use of x-rays or any other imaging techniques in the orthodontic treatment may be limited. Therefore, patient root geometries are not readily available for treatment planning, treatment diagnostics, and product development activities. Further, the detailed representation of the PDL in a PMM may lead to longer solve times and may impede solution convergence. In addition, as the PDL is fibrous in nature, its stress/strain response is non-linear and differs when the fibers are in tension or compression. Therefore, it may be important to include complex material models of the PDL for

finite-element analysis and simulations. This may further increase solve times and may impede solution convergence. Therefore, conventional finite element models generated by the FEMs typically treat the PDL as linear elastic and use a same proportionality factor (i.e., modulus) in tension and compression.

[0043] Further, tooth root and PDL geometries extracted from the x-ray or any other imaging techniques may contain inaccuracies. Inaccuracies in thickness profile of the detailed representation of the PDL, in particular, may have a huge impact on results. Moreover, the mechanical responses of the tooth for patients with different dental arrangements and conditions may also add large source of variability and unpredictability.

[0044] However, models for predicting the orthodontic biomechanics can still far outweigh accuracy challenges when considering other methods of predicting orthodontic biomechanics, such as trends, relative comparisons, and performance ranges.

[0045] Therefore, a practical and easy to implement representation is required for predicting the orthodontic biomechanics, such as the mechanical response of the PDL in orthodontic tooth movement simulations.

[0046] The present disclosure relates to a method for generating a design of an orthodontic appliance. The orthodontic appliance may be used in orthodontic treatments for moving one or more teeth of a patient from a malposition to a desired position in a dentition of the patient.

[0047] The method includes receiving a digital three-dimensional (3D) representation of at least one tooth. The method further includes accessing a Reduced Order Model (ROM) of a periodontal ligament (PDL) of the at least one tooth in the digital 3D representation. The ROM represents a mechanical response of the PDL of the at least one tooth. The method further includes generating the design of the orthodontic appliance based at least partially on the ROM of the PDL of the at least one tooth.

[0048] The method for generating the design of the orthodontic appliance based at least partially on the ROM of the PDL of the at least one tooth may be readily implemented in finite-element models. Further, the ROM of the PDL may accurately represent the mechanical response of the PDL to applied forces with reduced degrees of freedom (DOF) which may lead to about two orders of magnitude of reduction in simulation time. The ROM may retain critical information and relationships of the detailed representation of the PDL, and may further accurately represent the mechanical response of the PDL of the at least one tooth to the applied forces. In some cases, the simulation time may be reduced from hours to seconds.

[0049] Referring now to the Figures, FIG. 1 illustrates a digital three-dimensional (3D) representation **100** of a dental arch of a patient undergoing an orthodontic treatment, according to an embodiment of the present disclosure. It may be noted that the dental arch shown in FIG. 1 is a lower dental arch of the patient. In some other embodiments, the dental arch may include an upper dental arch of the patient. The dental arch includes a plurality of teeth **60**. In the illustrated embodiment of FIG. 1, the digital 3D representation **100** represents an initial arrangement of at least one tooth **62** from the plurality of teeth **60**. The at least one tooth **62** of the patient may be malpositioned. Therefore, the patient may be required to undergo the orthodontic treatment to correct malpositioning of the at least one tooth **62**. The at least one tooth **62** may be of the lower dental arch or the upper dental arch of the patient. Correcting the malpositioning of the at least one tooth **62** may improve facial appearance of the patient. Furthermore, correcting the malpositioning of the at least one tooth **62** may enhance function of the at least one tooth **62** by providing improved occlusion during mastication.

[0050] FIG. 2 shows an exemplary schematic sectional view of the at least one tooth **62** and a periodontal ligament (PDL) **64** of the at least one tooth **62**. The at least one tooth **62** may include one or more of a central incisor, a lateral incisor, a canine, a premolar, a first molar, a second molar, and a third molar. In the illustrated embodiment of FIG. 1, the at least one tooth **62** is a canine. Further, the at least one tooth **62** includes only one root **66**. In some other embodiments, the at least

one tooth **62** may include two or more roots **66**. For example, the at least one tooth **62** may include two or more roots **66** when the at least one tooth **62** is a molar (e.g., the first molar, the second molar, or the third molar).

[0051] FIG. **3** illustrates a schematic block diagram of a system **200** for generating a design of an orthodontic appliance (e.g., an orthodontic appliance **700** shown in FIG. **10**), according to an embodiment of the present disclosure. In some embodiments, the orthodontic appliance is at least one of an aligner, braces, or combinations thereof. In some other embodiments, the orthodontic appliance may be a retainer. The orthodontic appliance may be any appliance configured to apply forces and/or moments on the at least one tooth **62** (shown in FIG. **1**) to move the at least one tooth **62** from the initial arrangement (i.e., a malposition) shown in FIG. **1** to a target arrangement (i.e., a desired position) shown in FIG. **9** in a dentition of the patient.

[0052] The system **200** includes a processor **210**. The processor **210** is capable of executing one or more instructions. In some embodiments, the one or more instructions are stored in a memory. When the one or more instructions are executed by the processor **210**, the one or more instructions cause the processor **210** to perform one or more of the actions, operations, methods, or functions described herein.

[0053] In some embodiments, the processor **210** may include any suitable data processor for processing data. For example, the processor **210** may include a microprocessor, a microcontroller, a computer, or other suitable devices that control operation of devices and execute programs. Various other examples of the processor **210** include central processing units (“CPUs”), microcontrollers, programmable logic devices, field programmable gate arrays, digital signal processing (“DSP”) devices, and the like. The processor **210** may include any general variety device such as a reduced instruction set computing (“RISC”) device, a complex instruction set computing (“CISC”) device, or a specially designed processing device, such as an application-specific integrated circuit (“ASIC”) device.

[0054] In some embodiments, the processor **210** is configured to receive the digital 3D representation **100** of the at least one tooth **62**. The processor **210** can receive the digital 3D representation **100** locally or remotely via a network. In some embodiments, the processor **210** may be configured to receive the digital 3D representation **100** from the memory. In some embodiments, the processor **210** may be configured to receive the digital 3D representation **100** from an external database. In some embodiments, the processor **210** may be configured to receive the digital 3D representation **100** from the memory or the external database including the patient's file history or from a previous digital data capture. In some other embodiments, the processor **210** may be configured to retrieve the digital 3D representation **100** from a CAD file.

[0055] In some embodiments, dental practitioners may optically scan the at least one tooth **62** or the dental arch of the patient undergoing the orthodontic treatment to generate the digital 3D representation **100**. In some embodiments, the dental practitioners may perform an intraoral scan to generate the digital 3D representation **100**. In some embodiments, the dental practitioners may perform a digital data capture, a computed tomography (CT), or a computer-aided tomography (CAT) of the dental arch of the patient to generate the digital 3D representation **100**. In some other embodiments, the dental practitioners may indirectly perform a digital data capture of the dental arch of the patient by performing a digital data capture of a plaster model of the dental arch of the patient or of a dental impression of the dental arch of the patient, rather than directly capturing digital 3D representation **100** of the dental arch of the patient. In the case of using the dental impression, the digital data capture may be inverted from a negative volume to a positive volume. In some cases, the dental practitioners may send the digital 3D representation **100** to a manufacturing facility for generating the design of the orthodontic appliance and fabricating the orthodontic appliance.

[0056] In some embodiments, the system **200** may also include a display device **220** for displaying the digital 3D representation **100** of the at least one tooth **62** or the design of the orthodontic

appliance. The display device **220** can be implemented with any electronic display, for example a Cathode Ray Tube (CRT), a liquid crystal display (LCD), light emitting diode (LED) display, or organic light emitting diode (OLED) display.

[0057] In some embodiments, the system **200** further includes an input device **230** for receiving user commands or other information, for example to modify the design of the orthodontic appliance. The input device **230** can be implemented with any device for entering information or commands, for example a keyboard, a microphone, a cursor-control device, or a touch screen.

[0058] In some embodiments, the system **200** is in communication with a manufacturing device **240**. In some embodiments, the manufacturing system may be included in the system **200**. In some embodiments, the manufacturing device **240** may be a 3D printer.

[0059] In some embodiments, the manufacturing device **240** may be configured to fabricate the orthodontic appliance using at least one of an additive manufacturing process and a subtractive manufacturing process. The additive manufacturing process may include stereolithography (SLA) in which successive layers of material are laid down by the manufacturing device **240** under control of the processor **210** or a processor of the manufacturing device **240**. In other examples, the additive manufacturing process may include Fused Filament Fabrication (FFF). Powder Bed Fusion (PBF), and the like. The subtractive manufacturing process may include forming the orthodontic appliance from a blank by a milling process. The blank generally refers to a solid block of material from which the orthodontic appliance can be machined. In general, the blanks are attached to a support, a stub, or a mandrel that fits into the manufacturing device **240**. In some embodiments, the blank may have a rough shape of an outer contour of the orthodontic appliance.

[0060] In some embodiments, the manufacturing device **240** may be a thermoforming device configured for vacuum forming or positive pressure forming. In such a case, a model of a patient's tooth/teeth may be fabricated, for example, by an additive manufacturing process. Further, the model is used to fabricate the orthodontic appliance by means of heating a material the orthodontic appliance is to be made of and subsequently drawing the material over the model by means of a vacuum or positive pressure. In some embodiments, the material may be a transparent thermoplastic material. After the orthodontic appliance has been formed, the orthodontic appliance and the model can be separated from each other. Subsequently, the orthodontic appliance can be trimmed (optionally automated by CNC or robotic machinery, such as, end mill or LASER cutter) to remove excess material.

[0061] In some embodiments, the manufacturing device **240** may be any other manufacturing device configured to fabricate the orthodontic appliance based on the design by manufacturing processes, such as die pressing, slurry casting, injection molding, extrusion processes, rapid prototyping, and the like.

[0062] In some embodiments, the orthodontic appliance may be fabricated as a single part. In some other embodiments, different parts of the orthodontic appliance may be manufactured separately, and then assembled together by a suitable process.

[0063] In some embodiments, the manufacturing device **240** may also include an input device or an output device, such as a control input (e.g., button, touchpad, thumbwheel, etc.), or a display (e.g., LCD or LED display) to provide manufacturing status information.

[0064] In some other embodiments, the system **200** may not include the manufacturing device **240**. In such cases, the processor **210** may send the design of the orthodontic appliance generated by the system **200** to the manufacturing facility for fabricating the orthodontic appliance.

[0065] The system **200** can be implemented with, for example, a desktop, a notebook, or a tablet computer. The components of the system **200** may be combined, e.g., the tablet computer can incorporate the processor **210**, the display device **220**, and the input device **230** into a single unit.

[0066] FIG. 4 illustrates a schematic diagram of a Reduced Order Model (ROM) **300** of the PDL **64** (shown in FIG. 2) of the at least one tooth **62** (shown in FIG. 2), according to an embodiment of the present disclosure. The ROM **300** represents a mechanical response of the PDL **64** of the at

least one tooth **62**. Specifically, the ROM **300** represents the mechanical response of the PDL **64** of the at least one tooth **62** to applied forces (e.g., orthodontic force parameters **430** shown in FIG. 5). In some embodiments, the ROM **300** of the PDL **64** is independent of the applied forces.

[0067] The ROM **300** may help to accurately predict tooth displacements (e.g., tooth displacements **450** shown in FIGS. 6B to 6D) for the applied forces. As illustrated in FIG. 4, the ROM **300** may include a set of spring-like elements to represent the mechanical response of the PDL **64** of the at least one tooth **62**. The set of spring-like elements may include both axial and torsional springs.

[0068] In some embodiments, the ROM **300** includes a simplified mathematical representation. The simplified mathematical representation includes a plurality of numerical values representing compliance or stiffness of the PDL **64** of the at least one tooth **62** with respect to three translational degrees of freedom **302** and three rotational degrees of freedom **304**.

[0069] Referring to FIGS. 1 to 4, in some embodiments, the processor **210** is configured to access the ROM **300** of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. As discussed above, in some embodiments, the at least one tooth **62** includes the plurality of teeth **60** having different characteristics. For example, the plurality of teeth **60** may have different shapes, sizes, types, or number of roots.

[0070] In some embodiments, the processor **210** is configured to access the ROM **300** of the PDL **64** for each of the plurality of teeth **60**. In some embodiments, the processor **210** is configured to access the ROM **300** of the PDL **64** locally or remotely via the network.

[0071] In some embodiments, the processor **210** is further configured to generate the ROM **300** of the PDL **64** of the at least one tooth **62**. In some embodiments, the processor **210** is further configured to generate the ROM **300** of the PDL **64** of each of the plurality of teeth **60**.

[0072] In some embodiments, the processor **210** is configured to characterize a mechanical response of a baseline PDL of the at least one tooth **62** to the plurality of orthodontic force parameters (e.g., the orthodontic force parameters **430** shown in FIG. 5). The baseline PDL of the at least one tooth **62** has a baseline height **411** (shown in FIG. 5) and a baseline thickness **412** (shown in FIG. 5). In some embodiments, the baseline height **411** is a full height of the baseline PDL. In some embodiments, the baseline height **411** is about 16.3 millimeter (mm). In some embodiments, the baseline thickness **412** is an average thickness of the baseline PDL. In some embodiments, the baseline thickness **412** is about 0.245 mm. The baseline PDL may be used to calibrate the ROM **300**. The ROM **300** calibrated using the baseline PDL may be sufficiently accurate.

[0073] Further, in some embodiments, the processor **210** is configured to determine one or more ROM parameters of the ROM **300** of the baseline PDL based at least on the digital 3D representation **100**. In some embodiments, the one or more ROM parameters of the ROM **300** include, but are not limited to, one or more stiffness parameters or one or more compliance parameters.

[0074] In some embodiments, one or more ROM parameters are determined from experimental measurements of the relationship between forces and displacements of actual teeth in human or animal subjects.

[0075] In some embodiments, one or more ROM parameters are determined by finding the best fit between tooth movement predicted using a PMM and observed movement of at least one tooth during at least a portion of orthodontic treatment conducted on at least one patient.

[0076] FIG. 5 illustrates a schematic diagram of a physics-based mathematical model (PMM) **400** of the at least one tooth **62** (shown in FIG. 2), according to an embodiment of the present disclosure. The PMM **400** is based on the digital 3D representation **100** (shown in FIG. 1). Specifically, the PMM **400** of the at least one tooth **62** is based on the digital 3D representation **100**. The PMM **400** includes at least a PDL model **410**. In some embodiments, the PMM **400** may further include a tooth model **418**.

[0077] In some embodiments, the processor **210** (shown in FIG. 3) is configured to generate the

PMM **400** based on the digital 3D representation **100**. In some embodiments, the processor **210** is configured to generate the PDL model **410** of the at least one tooth **62** based on non-linear material characteristics of the baseline PDL of the at least one tooth **62**. In some embodiments, hyperelastic material models may be used to describe the non-linear material characteristics of the baseline PDL of the at least one tooth **62**. The hyperelastic material models may capture PDL compression and tension behavior accurately.

[0078] In some embodiments, the non-linear material characteristics of the baseline PDL are according to Equation 1 provided below:

$$[00001] \ U = \text{.Math.} \prod_{i=1}^N \frac{2}{\lambda_i} \left[\lambda_{1,\text{sub},i}^i + \lambda_{2,\text{sub},i}^i + \lambda_{3,\text{sub},i}^i - 3 + \frac{1}{J^{\text{el}}} (J^{\text{el}})^{\frac{1}{\alpha_{\text{sub},i}} - 1} \right] \quad (1) \quad [0079] \text{ where, } [0080]$$

$\lambda_{\text{sub},1}$, $\lambda_{\text{sub},2}$, $\lambda_{\text{sub},3}$ are principal stretches, [0081] J^{el} is a volumetric stretch, [0082] $m_{\text{sub},i}$, $\alpha_{\text{sub},i}$, $\beta_{\text{sub},i}$, are fitted parameters, and [0083] $i=1, 2 \dots N$.

[0084] Further, the at least one tooth **62** and its jaw bone can be approximated as rigid bodies.

Therefore, in some embodiments, the processor **210** is configured to generate the tooth model **418** of the at least one tooth **62** based on rigid material characteristics of the at least one tooth **62**.

[0085] In some embodiments, the ROM **300** (shown in FIG. **4**) is determined based on the PMM **400**. In such embodiments, the processor **210** is configured to determine, for the at least one tooth **62**, a plurality of tooth movement parameters (e.g., a tooth movement parameter **440** shown in FIGS. **6B-6C**) corresponding to the plurality of orthodontic force parameters **430** applied at a reference point **420** disposed on or within the at least one tooth **62** based on the PMM **400**. In other words, the PMM **400** is simulated a plurality of times by applying the plurality of orthodontic force parameters **430** at the reference point **420** to determine the plurality of tooth movement parameters corresponding to the plurality of orthodontic force parameters **430** applied at the reference point **420**. Each of the plurality of orthodontic force parameters **430** may include a combination of axial and torsional forces, such as F_x , F_y , F_z , M_x , M_y , and M_z . The plurality of orthodontic force parameters **430** may include different combinations of the orthodontic force parameters **430**. The location of the reference point **420** is arbitrary and may be defined on the surface or within the volume of the at least one tooth **62**. In some embodiments, the reference point is located in the center of the root volume near an expected location of the center of resistance.

[0086] FIG. **6A** illustrates a schematic isometric view of the PMM **400** and an orthodontic force parameter **430** from the plurality of orthodontic force parameters **430** applied at the reference point **420**. FIG. **6B** illustrates a schematic side view of the PMM **400** upon application of the orthodontic force parameter **430** at the reference point **420**. FIG. **6C** illustrates a schematic front view of the PMM **400** upon application of the orthodontic force parameter **430** at the reference point **420**. FIG. **6D** illustrates a schematic top view of the PMM **400** upon application of the orthodontic force parameter **430** at the reference point **420**.

[0087] Referring to FIGS. **4**, **5**, and **6A-6C**. in some embodiments, each of the plurality of tooth movement parameters **440** includes a plurality of tooth displacements **450** at the reference point **420** due to an application of a corresponding orthodontic force parameter **430** from the plurality of orthodontic force parameters **430** at the reference point **420**. In some embodiments, the plurality of tooth displacements **450** may include a plurality of axial and torsional displacements, such as U_x , U_y , U_z , U_{R_x} , U_{R_y} , and U_{R_z} (collectively referred as “the plurality of tooth displacements U ”) due to the application of the corresponding orthodontic force parameter F_x , F_y , F_z , M_x , M_y , and M_z (collectively referred as “the plurality of orthodontic force parameters F ”) from the plurality of orthodontic force parameters **430** at the reference point **420**.

[0088] In some embodiments, the processor **210** (shown in FIG. **3**) is configured to determine a calibrated compliance matrix CRP of the baseline PDL of the at least one tooth **62** (shown in FIG. **2**) based at least on the plurality of tooth movement parameters **440** (i.e., the plurality of tooth movement parameters U in Equation 2 provided below) and the plurality of orthodontic force parameters **430** (i.e., the plurality of orthodontic force parameters F in Equation 2 provided below).

In some embodiments, the calibrated compliance matrix CRP provides a correlation between the plurality of orthodontic force parameters **430** and the plurality of tooth movement parameters **440**. In some embodiments, the calibrated compliance matrix CRP is according to Equation 2 provided below:

$$[00002] \quad U = C_{RP} F = \begin{pmatrix} c_{11} & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ c_{12} & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ c_{13} & c_{23} & c_{33} & c_{34} & c_{35} & c_{36} \\ c_{14} & c_{24} & c_{34} & c_{44} & c_{45} & c_{46} \\ c_{15} & c_{25} & c_{35} & c_{45} & c_{55} & c_{56} \\ c_{16} & c_{26} & c_{36} & c_{46} & c_{56} & c_{66} \end{pmatrix} \begin{Bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{Bmatrix} \quad (2)$$

[0089] In some embodiments, the processor **210** is further configured to determine a plurality of center of resistance parameters X.sub.CoR, Y.sub.CoR, Z.sub.CoR of the baseline PDL of the at least one tooth **62** based on the calibrated compliance matrix CRP. In some embodiments, the processor **210** is configured to determine the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR based on the calibrated compliance matrix C.sub.RP through kinematic analysis of the plurality of tooth movement parameters **440**. In some embodiments, the plurality of center of resistance parameters X.sub.CoR, Y.sub.CoR, Z.sub.CoR based on the calibrated compliance matrix .sub.CRP are determined based on Equation 3 provided below:

$$[00003] \quad \begin{aligned} x_{CoR} &= -C_{35} / C_{55} \\ y_{CoR} &= C_{34} / C_{44} \\ z_{CoR} &= C_{15} / C_{55} \end{aligned} \quad (3)$$

[0090] In some embodiments, the processor **210** is further configured to determine a reference mapping matrix Γ based on the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR of the baseline PDL.

[0091] In some embodiments, the reference mapping matrix Γ is according to Equation 4 provided below:

$$[00004] \quad = \begin{pmatrix} 1 & 0 & 0 & 0 & -r_z & r_y \\ 0 & 1 & 0 & r_z & 0 & -r_x \\ 0 & 0 & 1 & -r_y & r_x & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (4) \quad [0092] \text{ where, } [0093] \text{ } r_x, r_y \text{ and } r_z \text{ are the distances}$$

between the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR and the reference point **420**, respectively.

[0094] In some embodiments, the processor **210** is configured to determine a characteristic compliance matrix C.sub.COR of the baseline PDL of the at least one tooth **62** based on the calibrated compliance matrix CRP and the reference mapping matrix Γ .

[0095] In some embodiments, the characteristic compliance matrix C.sub.COR represents a correlation between orthodontic force parameters applied at a center of resistance (CoR) **405** with one or more displacements at the CoR **405**.

[0096] In some embodiments, the characteristic compliance matrix C.sub.COR is determined according to Equation 5 provided below:

$$[00005] \quad C_{CoR} = {}^{-1} C_{RP} {}^{-T} \quad (5)$$

[0097] In some embodiments, the characteristic compliance matrix C.sub.COR is a diagonal matrix.

[0098] In some embodiments, the processor **210** is configured to determine a plurality of numerical

values representing compliance of the baseline PDL based on the characteristic compliance matrix C.sub.COR. For example, the plurality of numerical values representing compliance of the baseline PDL may be indicative of spring constants of the spring-like elements of the ROM **300** that represent the mechanical response of the PDL **64** of the at least one tooth **62**. Specifically, the plurality of numerical values representing compliance of the baseline PDL may be converted through matrix inverse operation to the spring constants of the spring-like elements of the ROM **300** that represent the mechanical response of the PDL **64** of the at least one tooth **62**. In some embodiments, the plurality of numerical values representing compliance of the baseline PDL can be directly obtained from the characteristic compliance matrix C.sub.COR.

[0099] In some other embodiments, the processor **210** is configured to determine a stiffness matrix as an inverse of the characteristic compliance matrix C.sub.COR. In such embodiments, the processor **210** is configured to determine the plurality of numerical values representing stiffness of the baseline PDL based on the stiffness matrix. For example, the plurality of numerical values representing stiffness of the baseline PDL may be the spring constants of the spring-like elements of the ROM **300** that represent the mechanical response of the PDL **64** of the at least one tooth **62**.

[0100] In some embodiments, the processor **210** is configured to determine the simplified mathematical representation of the baseline PDL of the at least one tooth **62** based on the plurality of numerical values. In some embodiments, the simplified mathematical representation of the baseline PDL of the at least one tooth **62** based on the plurality of numerical values is according to the Equation 6 provided below:

$$[00006] \quad U = C_{CoR} F = \begin{pmatrix} c11 & 0 & 0 & 0 & 0 & 0 \\ 0 & c22 & 0 & 0 & 0 & 0 \\ 0 & 0 & c33 & 0 & 0 & 0 \\ 0 & 0 & 0 & 11 & 0 & 0 \\ 0 & 0 & 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 & 0 & 33 \end{pmatrix} \begin{Bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{Bmatrix} \quad (6)$$

where, c11, c22, c33, β_{11} , 22, β_{33} are the plurality of numerical values representing compliance of the baseline PDL.

[0101] The simplified mathematical representation of the baseline PDL of the at least one tooth **62** can be used for orthodontic treatment planning, orthodontic treatment diagnostics, and development of the orthodontic appliance **700**.

[0102] In some embodiments, the processor **210** is configured to determine the ROM **300** based on the simplified mathematical representation of the baseline PDL of the at least one tooth **62**.

[0103] FIG. 7 illustrates a schematic diagram of an updated PMM **500** of the at least one tooth **62** (shown in FIG. 2), according to an embodiment of the present disclosure. The updated PMM **500** of the at least one tooth **62** includes the ROM **300** of the PDL **64** (shown in FIG. 2) of the at least one tooth **62**. Specifically, the updated PMM **500** of the at least one tooth **62** includes the ROM **300** of the PDL **64** of the at least one tooth **62** at the CoR **405**.

[0104] Referring to FIGS. 1 to 7, in some embodiments, the processor **210** is configured to remove the PDL model **410** from the PMM **400** and apply the ROM **300** at the CoR **405** to generate the updated PMM **500** of the at least one tooth **62**. In some embodiments, the processor **210** is configured to remove the PDL model **410** from the PMM **400** and apply the ROM **300** consisting of the set of spring-like elements to represent the mechanical response of the PDL **64** of the at least one tooth **62** at the CoR **405** to generate the updated PMM **500** of the at least one tooth **62**.

Applying the ROM **300** at the CoR **405** to generate the updated PMM **500** of the at least one tooth **62** may simplify the PDL model **410** and thereby reduce computation time required to predict the tooth displacements U from hours to seconds. In other words, the updated PMM **500** may require much lesser computation time as compared to computation time required by the PMM **400** of the at least one tooth **62** to predict the tooth displacements U.

[0105] Since the ROM **300** is calibrated using simulation results where the non-linear material characteristics of the PDL **64** are modeled realistically, the ROM **300** captures the mechanical response of the PDL **64** of the at least one tooth **62** by considering the non-linear material characteristics of the PDL **64**. This may further ensure that the ROM **300** is accurate.

[0106] FIG. **8** illustrates a schematic diagram of the updated PMM **500** of the plurality of teeth **60** (shown in FIG. **1**), according to another embodiment of the present disclosure. In the illustrated embodiment of FIG. **8**, the plurality of teeth **60** has different characteristics. In some embodiments, the processor **210** (shown in FIG. **3**) is configured to remove the PDL model **410** (shown in FIG. **5**) from the PMM **400** for each of the plurality of teeth **60** and apply the ROM **300** of the PDL **64** of each of the plurality of teeth **60** at the CoR **405** of each of the plurality of teeth **60** to generate the updated PMM **500**.

[0107] FIG. **9** illustrates a schematic diagram of a design **600** of the dental arch of the patient undergoing the orthodontic treatment, according to an embodiment of the present disclosure. In the illustrated embodiment of FIG. **9**, the design **600** represents the target arrangement of at least one tooth **62** from the plurality of teeth **60**.

[0108] Referring to FIGS. **1** to **9**, in some embodiments, the processor **210** is further configured to receive the target arrangement of the at least one tooth **62**. In some embodiments, the processor **210** is further configured to determine one or more force parameters (e.g., the orthodontic force parameters **F**) based on the updated PMM **500** to achieve the target arrangement from the initial arrangement of the at least one tooth **62**. In some embodiments, the processor **210** may be configured to determine the one or more force parameters based on the simplified mathematical representation (i.e., Equation 6 provided above) of the baseline PDL of the at least one tooth **62**.

[0109] FIG. **10** illustrates a schematic diagram of the design of the orthodontic appliance **700** and the dental arch of the patient, according to an embodiment of the present disclosure. The orthodontic appliance **700** applies the one or more force parameters to the at least one tooth **62** to achieve the target arrangement. In some embodiments, the processor **210** (shown in FIG. **3**) is configured to generate the design of the orthodontic appliance **700**, such that the orthodontic appliance **700** applies the one or more force parameters to the at least one tooth **62** to achieve the target arrangement.

[0110] In some embodiments, the orthodontic appliance is designed using an inverse method. The method is shown in FIGS. **17A-17J**, which depict certain elements in representative fashion and do not depict exact anatomical or appliance geometries. The teeth are represented as ovals, viewed in a direction generally perpendicular to the occlusal plane. In such embodiments, the method comprises some or all of the following steps: [0111] i. Receiving a digital three-dimensional (3D) representation **900** of a patient's dentition comprising first tooth positions (FIG. **17A**); [0112] ii. Determining desired second positions of at least one tooth **904** (FIG. **17B**) and creating a modified 3D representation **902** of the patient's dentition with at least one tooth **904** in the desired second position; [0113] iii. Building a 3D model of an orthodontic appliance **906** corresponding to the second tooth positions (FIG. **17C**); [0114] iv. Generating a physics-based mathematical model (PMM) by combining the modified 3D representation of the patient's dentition **904** and the 3D model of the orthodontic appliance **906** (FIG. **17C**); [0115] v. Identifying at least one anticipated contact area **908** between the appliance **910** and the at least one tooth **904**, based on the intersection between the first tooth positions and the shape of the appliance **910** that corresponds to the second tooth positions; [0116] vi. Calculating the forces **912**, **914** required to move at least one tooth from its first position to its second position, when applied at the likely contact areas, using a Reduced Order Model (ROM) of the periodontal ligament (PDL) of the at least one tooth (FIG. **17D**); [0117] vii. Modifying the PMM by removing the at least one tooth and applying the opposite of the required forces **916**, **918** to move the at least one tooth from its first position to its second position at the at least one likely contact area **908** between the appliance **910** and at least one tooth (FIG. **17E**); [0118] viii. Simulating deformation **920** of the orthodontic appliance **910** due to the applied

forces **919** and interactions between the appliance **910** and any teeth **901** included in the PMM (FIG. **17F**); [0119] ix. Calculating discrepancies **924** between the deformed appliance **922** and the 3D model of the orthodontic appliance **910** corresponding to the second tooth positions (FIG. **17G**); [0120] X. Updating the 3D model of the orthodontic appliance **910** to account for the discrepancies to create a discrepancy updated model **930** (FIG. **17H**); [0121] xi. Repeating steps v. through ix. until the discrepancies between the deformed appliance shape solution and the target deformed appliance shape are less than a maximum allowed discrepancy to create a final updated model of the appliance (**940**); [0122] xii. Generating a validation PMM by combining the original 3D representation of the patient's dentition **900** and the final updated 3D model of the orthodontic appliance **940** (FIG. **17I**); and [0123] xiii. Solving/simulating the validation PMM and comparing the resulting predicted tooth positions (**950**) with the desired second tooth positions to validate or reject the final orthodontic appliance design (**960**) corresponding to the final updated 3D model of the orthodontic appliance (Fig, **17J**).

[0124] In some embodiments the inverse method comprises a finite-element model. In some embodiments, the inverse method comprises an interference fit method to simulate interactions between the orthodontic appliance and any teeth included in the PMM. In some embodiments, the inverse method comprises simulating dynamic interactions between the orthodontic appliance and any teeth included in the PMM while the appliance is displaced toward and seated on the teeth.

[0125] Therefore, the system **200** may provide various insights, such as the one or more force parameters required to achieve the target arrangement, the one or more force parameters applied by the orthodontic appliance **700**, a placement or an orientation of the orthodontic appliance **700** required to achieve the one or more force parameters, a time period required to achieve the target arrangement, and so forth. These insights may further improve the orthodontic treatment planning, the orthodontic treatment diagnostics, and the development of the orthodontic appliance **700**.

[0126] FIG. **11** illustrates a schematic view of the updated PMM **500** and a virtual orthodontic appliance design **750**. according to an embodiment of the present disclosure.

[0127] Referring to FIGS. **1** to **11**, in some embodiments, the processor **210** is configured to receive an orthodontic appliance data **702** (schematically shown in FIG. **3**). The orthodontic appliance data **702** includes at least one of orthodontic appliance material properties and orthodontic appliance geometry. In some embodiments, the processor **210** is configured to form the virtual orthodontic appliance design **750** based on the orthodontic appliance data **702**.

[0128] In some embodiments, the processor **210** is configured to modify the updated PMM **500** of the at least one tooth **62** by adding the virtual orthodontic appliance design **750** to the updated PMM **500**.

[0129] FIG. **12** illustrates a schematic view of a design **650** of the dental arch of the patient, according to an embodiment of the present disclosure.

[0130] Referring to FIGS. **11** and **12**, the design **650** illustrates the at least one tooth **62** in a resulting tooth arrangement due to an interaction between the at least one tooth **62** in the updated PMM **500** and the virtual orthodontic appliance design **750**.

[0131] In some embodiments, the processor **210** (shown in FIG. **3**) is configured to simulate the interaction between the at least one tooth **62** in the updated PMM **500** and the virtual orthodontic appliance design **750** and determine the resulting tooth arrangement.

[0132] In some embodiments, the processor **210** is further configured to compare the resulting tooth arrangement from the simulated interaction with the target arrangement.

[0133] Therefore, in some embodiments, the processor **210** may be configured to verify that the virtual orthodontic appliance design **750** is applying the one or more force parameters to achieve the target arrangement of the at least one tooth **62** based on the comparison.

[0134] In some embodiments, the system **200** may be configured to display the virtual orthodontic appliance design **750**, via the display device **220**, and modify the virtual orthodontic appliance design **750**.

[0135] In some other embodiments, the processor **210** may execute one or more software applications to automatically modify the virtual orthodontic appliance design **750** if the virtual orthodontic appliance design **750** is not applying the one or more force parameters required to achieve the target arrangement.

[0136] In some other embodiments, the dental practitioner may manually modify the virtual orthodontic appliance design **750** if the virtual orthodontic appliance design **750** is not applying the one or more force parameters required to achieve the target arrangement via the input device **230** (shown in FIG. 3).

[0137] In some embodiments, the manufacturing device **240** (shown in FIG. 3) is configured to fabricate the orthodontic appliance **700** based on the virtual orthodontic appliance design **750**. Specifically, the manufacturing device **240** may be configured to fabricate the orthodontic appliance **700** based on the virtual orthodontic appliance design **750** if the virtual orthodontic appliance design **750** is applying the one or more force parameters required to achieve the target arrangement.

[0138] Therefore, the system **200** may improve development of the virtual orthodontic appliance design **750** by modifying the orthodontic appliance data **702** (i.e., at least one of the orthodontic appliance material properties and the orthodontic appliance geometry) to fabricate the orthodontic appliance **700**, such that the one or more force parameters required to achieve the target arrangement are applied by the orthodontic appliance **700**. This may further improve the orthodontic treatment planning and the orthodontic treatment diagnostics.

[0139] FIG. 13 illustrates a schematic view of a PMM **401**, according to another embodiment of the present disclosure. The PMM **401** is substantially similar to the PMM **400** shown in FIG. 5. However, the PMM **401** includes a PDL model **413** of the PDL **64** (shown in FIG. 2) of the at least one tooth **62** (shown in FIG. 2). In this embodiment, the PDL **64** of the at least one tooth **62** has a height **414**. The height **414** is different from the baseline height **411** (shown in FIG. 5) of the baseline PDL of a similar tooth, i.e., the at least one tooth **62**.

[0140] The PDL **64** having the height **414** different from the baseline height **411** of the baseline PDL of the similar tooth may have different at least one center of resistance parameter $X_{sub.CoR}$, $V_{sub.CoR}$, $Z_{sub.CoR}$. This may be due to a differing constraint to the root **66** (shown in FIG. 2) of the at least one tooth **62** by the PDL **64**. The at least one center of resistance parameter $X_{sub.CoR}$, $Y_{sub.CoR}$, $Z_{sub.CoR}$ may be linearly proportional to the height **414** of the PDL **64**. Specifically, the center of resistance parameter $Z_{sub.CoR}$ may be linearly proportional to the height **414** of the PDL **64**.

[0141] Further, since a resistance to the tooth displacements **450** (shown in FIGS. 6B to 6D) may be different due to the differing constraint to the root **66**, and as the PDL **64** having the height **414** different from the baseline height **411** may have a different volume to store a strain energy caused by the tooth displacements **450**, there may be an increase in compliance of the PDL **64**.

[0142] Therefore, the PDL **64** having the height **414** different from the baseline height **411** may have a different calibrated compliance matrix from the calibrated compliance matrix $C_{sub.RP}$ and a different plurality of numerical values from the plurality of numerical values representing compliance of the baseline PDL.

[0143] Referring to FIGS. 1 to 5 and 13, in some embodiments, the processor **210** is configured to scale the ROM **300** of the PDL **64** of the at least one tooth **62**.

[0144] In some embodiments, the processor **210** is configured to determine the height **414** of the PDL **64** of the at least one tooth **62** based on the digital 3D representation **100**.

[0145] In some embodiments, the processor **210** is configured to determine at least one corrected center of resistance parameter different from corresponding at least one center of resistance parameter (e.g., the center of resistance parameter $X_{sub.CoR}$) from the plurality of center of resistance parameters $X_{sub.CoR}$, $V_{sub.CoR}$, $Z_{sub.CoR}$ of the baseline PDL.

[0146] In some embodiments, the processor **210** is configured to determine a centroid **406** of the

PDL **64** of the at least one tooth **62** from the digital 3D representation **100** (shown in FIG. 3). The centroid **406** corresponds to the at least one corrected center of resistance parameter. In some embodiments, the centroid **406** corresponds to a corrected CoR having the at least one corrected center of resistance parameter.

[0147] In some embodiments, the processor **210** is configured to determine a corrected reference mapping matrix Γ_s based on the at least one corrected center of resistance parameter (i.e., Z.sub.CoR) using Equation 4 in order to scale the ROM **300** of the PDL **64** of the at least one tooth **62** from that of the baseline PDL.

[0148] In some embodiments, the processor **210** is further configured to determine a first scale factor $f_{\text{sub.A}}$ of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. In some embodiments, the first scale factor $f_{\text{sub.A}}$ is a combination of the height **414** of the PDL **64** and the baseline height **411** (shown in FIG. 5) of the baseline PDL.

[0149] In some embodiments, the first scale factor $f_{\text{sub.A}}$ is according to Equation 7 provided below:

[00007] $f_A = \left(\frac{h_b}{h_s}\right)^2$ (7) [0150] where, [0151] $h_{\text{sub.s}}$ is the height **414** of the PDL **64**, and [0152] $h_{\text{sub.b}}$ is the baseline height **411**.

[0153] In some embodiments, a combination of the first scale factor $f_{\text{sub.A}}$ and the characteristic compliance matrix $C_{\text{sub.COR}}$ of the baseline PDL associates a corrected characteristic compliance matrix $C_{\text{sub.CORH}}$ of the PDL with the height **414** of the PDL **64**.

[0154] In some embodiments, the corrected characteristic compliance matrix $C_{\text{sub.CORH}}$ is obtained using Equation 8 provided below:

[00008] $C_{\text{CORH}} = f_A C_{\text{COR}}$ (8) [0155] where, [0156] $C_{\text{sub.CORH}}$ is a characteristic compliance matrix for the PDL **64** having the height **414**.

[0157] In some embodiments, the $C_{\text{sub.CORH}}$ may be obtained similarly as the characteristic compliance matrix $C_{\text{sub.COR}}$ of the baseline PDL except the PDL model **410** has the height **414** instead of the baseline height **411**.

[0158] In some embodiments, a characteristic compliance matrix ratio x is according to Equation 9 provided below:

[00009] $x = C_{\text{CORH}}(C_{\text{COR}})^{-1}$ (9)

[0159] The characteristic compliance matrix ratio x is indicative of effects of height variation of the PDL **64** from the baseline PDL.

[0160] FIG. **14** illustrates the 2D image **101** of the PDL **64** of the at least one tooth **62** (shown in FIG. 2). In some embodiments, the 2D image **101** may be obtained from a 2D x-ray image without a need of digital 3D representation **100**. In some other embodiments, the processor **210** (shown in FIG. 3) is configured to create the 2D image **101** of the PDL **64** of the at least one tooth **62** from the digital 3D representation **100** (shown in FIG. 1). The 2D image **101** of the PDL **64** is along a height axis (i.e., the z-axis) extending along the height **414** of the PDL **64** and an axis (i.e., the y-axis) orthogonal to the height axis.

[0161] In some embodiments, the processor **210** is configured to determine a plurality of points **105** on the PDL **64** of the at least one tooth **62** from the 2D image **101**. In the illustrated embodiment of FIG. **14**, the plurality of points **105** includes a top leftmost point on the PDL **64**, a top rightmost point on the PDL **64**, and a bottommost point on the PDL **64**. Therefore, the plurality of points **105** includes three points. However, in some other embodiments, the plurality of points **105** may include any number of points based on desired application attributes.

[0162] In some embodiments, the processor **210** is configured to connect the plurality of points **105** to construct a triangle **106** or a semi-ellipse **107**.

[0163] In some embodiments, the processor **210** is configured to determine an area centroid **407A** of the triangle **106** or an area centroid **407B** the semi-ellipse **107**. The area centroids **407A**, **407B**

are collectively referred to as the area centroid **407**. The area centroid **407** corresponds to the at least one corrected center of resistance parameter. In some embodiments, the area centroid **407** corresponds to the corrected CoR having the at least one corrected center of resistance parameter. [0164] FIG. **15** illustrates a schematic view of a PMM **402**, according to another embodiment of the present disclosure. The PMM **402** is substantially similar to the PMM **400** shown in FIG. **5**. However, the PMM **402** includes a PDL model **415** of the PDL **64** (shown in FIG. **2**) of the at least one tooth **62** (shown in FIG. **2**). In this embodiment, the PDL **64** of the at least one tooth **62** has a thickness **416**. The thickness **416** is different from the baseline thickness **412** (shown in FIG. **5**) of the baseline PDL of the similar tooth, i.e., the at least one tooth **62**.

[0165] The PDL **64** having the thickness **416** different from the baseline thickness **412** of the baseline PDL of the similar tooth may have a differing constraint to the root **66** (shown in FIG. **2**) of the at least one tooth **62** by the PDL **64**. For example, a thicker PDL may have lesser constraint to the root **66** of the at least one tooth **62** by the PDL **64**. Thus, there may be an increase in compliance of the PDL **64**.

[0166] Therefore, the PDL **64** having the thickness **416** different from the baseline thickness **412** may have a different calibrated compliance matrix from the calibrated compliance matrix C.sub.RP and a different plurality of numerical values from the plurality of numerical values representing compliance of the baseline PDL.

[0167] Referring to FIGS. **1** to **5** and **15**, in some embodiments, the processor **210** is configured to determine the thickness **416** of the PDL **64** of the at least one tooth **62** based on the digital 3D representation **100**.

[0168] In some embodiments, the processor **210** is configured to determine the at least one corrected center of resistance parameter (e.g., at least one of the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR) different from corresponding at least one center of resistance parameter from the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR of the baseline PDL.

[0169] In some embodiments, the processor **210** is also configured to determine a corrected reference mapping matrix Γ_t based on the at least one corrected center of resistance parameter in order to map the one or more displacements at the center of resistance **405** of the at least one tooth **62** to the other locations of the at least one tooth **62**.

[0170] In some embodiments, the processor **210** is further configured to determine a second scale factor f.sub.B of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. In some embodiments, the second scale factor f.sub.B is a combination of the thickness **416** of the PDL **64** and the baseline thickness **412** of the baseline PDL.

[0171] In some embodiments, the second scale factor f.sub.B is according to Equation 10 provided below:

[00010] $f_B = \frac{t}{t_b}$ (10) [0172] where, [0173] t is the thickness **416** of the PDL **64**, and [0174] t.sub.b is the baseline thickness **412**.

[0175] In some embodiments, a combination of the second scale factor f.sub.B and the characteristic compliance matrix C.sub.COR of the baseline PDL associates a corrected characteristic compliance matrix C.sub.CORT of the PDL **64** with the thickness **416** of the PDL **64**.

[0176] In some embodiments, the corrected characteristic compliance matrix C.sub.CORT is obtained using Equation 11 provided below:

[00011] $C_{CORT} = f_B C_{COR}$ (11)

[0177] Referring to FIGS. **1** to **5** and **13** to **15**, in some embodiments, the height **414** and the thickness **416** are different from the baseline height **411** and the baseline thickness **412**, respectively, of the baseline PDL of the similar tooth.

[0178] In some embodiments, the processor **210** is configured to determine the height **414** and the thickness **416** of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. In

some embodiments, the processor **210** is configured to determine the height **414** and the thickness **416** of the PDL **64** of the at least one tooth **62** based on the digital 3D representation **100**.

[0179] In some embodiments, the processor **210** is configured to determine the at least one corrected center of resistance parameter different from corresponding at least one center of resistance parameter from the plurality of center of resistance parameters X.sub.CoR, Y.sub.CoR, Z.sub.CoR of the baseline PDL. In some embodiments, the at least one corrected center of resistance parameter may be obtained by determining the centroid **406** and/or the area centroid **407**.

[0180] In some embodiments, the processor **210** is also configured to determine a corrected reference mapping matrix Γ_{st} based on the at least one corrected center of resistance parameter in order to map the one or more displacements at the center of resistance **405** of the at least one tooth **62** to the other locations of the at least one tooth **62**.

[0181] In some embodiments, the processor **210** is further configured to determine a combined scale factor f.sub.AB of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. In some embodiments, the combined scale factor f.sub.AB is a combination of the first scale factor f.sub.A and the second scale factor f.sub.B.

[0182] In some embodiments, a combination of the combined scale factor f.sub.AB and the characteristic compliance matrix C.sub.COR of the baseline PDL associates a corrected characteristic matrix C.sub.CORHT of the PDL **64** with the height **414** and the thickness **416** of the PDL **64**. In some embodiments, the corrected characteristic compliance matrix C.sub.CORHT is obtained using Equation 12 and Equation 13 provided below:

[00012] $C_{CORHT} = f_{AB} C_{COR}$ (12) $f_{AB} = f_A f_B$ (13) [0183] where. [0184] f.sub.A and f.sub.B are defined in Equation 7 and Equation 10, respectively.

[0185] Therefore, the ROM **300** of the baseline PDL may be modified to reflect the specific variations (i.e., the height **414** and the thickness **416**) in PDL shapes of different patients. The ROM **300** of the PDL **64** may accurately represent the mechanical response of the PDL **64**. Since the ROM **300** of the baseline PDL may be modified using the first scale factor f.sub.A, the second scale factor f.sub.B, and/or the combined scale factor f.sub.AB to reflect the specific variations in PDL shapes of the patients, there may not be a need to develop a ROM for each specific patient.

[0186] FIG. **16** is a flowchart illustrating a method **800** for generating the design of the orthodontic appliance **700**. according to an embodiment of the present disclosure. The orthodontic appliance **700** is at least one of an aligner, braces, or combinations thereof.

[0187] The method **800** will be described with reference to FIGS. **1** to **15**. The method **800** includes the following steps:

[0188] At step **802**, the method **800** includes receiving the digital 3D representation **100** of the at least one tooth **62**. In some embodiments, the method includes receiving the digital 3D representation **100** of the at least one tooth **62** by the processor **210**.

[0189] At step **804**, the method **800** includes accessing the ROM **300** of the PDL **54** of the at least one tooth **62** in the digital 3D representation **100**. In some embodiments, the at least one tooth **62** includes the plurality of teeth **60** having the different characteristics. In some embodiments, the method **800** further includes accessing the ROM **300** of the PDL **64** for each of the plurality of teeth **60**.

[0190] As discussed above, the ROM **300** represents the mechanical response of the PDL **54** of the at least one tooth **62**. The ROM **300** includes the simplified mathematical representation including the plurality of numerical values representing compliance or stiffness of the PDL **64** of the at least one tooth **62** with respect to the three translational degrees of freedom **302** and the three rotational degrees of freedom **304**.

[0191] In some embodiments, accessing the ROM **300** includes generating the ROM **300**. In some embodiments, accessing the ROM **300** for each of the plurality of teeth **60** includes generating the ROM **300** for each of the plurality of teeth **60**.

[0192] Further, in some embodiments, generating the ROM **300** includes characterizing the mechanical response of the baseline PDL of the at least one tooth **62** to the plurality of orthodontic force parameters **430** and determining the one or more ROM parameters of the ROM **300** of the baseline PDL based at least on the digital 3D representation **100**.

[0193] In some embodiments, the method **800** further includes generating the PMM **400** of the at least one tooth **62** based on the digital 3D representation **100**. The PMM **400** includes at least the PDL model **410**. In some embodiments, generating the PMM **400** further includes generating the PDL model **410** of the at least one tooth **62** based on the non-linear material characteristics of the baseline PDL of the at least one tooth **62**.

[0194] In some embodiments, the ROM **300** is determined further based on the PMM **400**.

[0195] In some embodiments, generating the ROM **300** further includes determining, for the at least one tooth **62**, the plurality of tooth movement parameters **440** corresponding to the plurality of orthodontic force parameters **430** applied at the reference point **420** disposed on the at least one tooth **62** based on the PMM **400**. In some embodiments, each of the plurality of tooth movement parameters **440** includes the plurality of tooth displacements **450** at the reference point **420** due to the application of the corresponding orthodontic force parameter **430** from the plurality of orthodontic force parameters **430** at the reference point **420**.

[0196] In some embodiments, generating the ROM **300** further includes determining the calibrated compliance matrix C.sub.RP of the baseline PDL of the at least one tooth **62** based at least on the plurality of tooth movement parameters **440** and the plurality of orthodontic force parameters **430**.

[0197] In some embodiments, the calibrated compliance matrix C.sub.RP provides the correlation between the plurality of orthodontic force parameters **430** and the plurality of tooth movement parameters **440**.

[0198] In some embodiments, generating the ROM **300** further includes determining the plurality of center of resistance parameters X.sub.CoR, Y.sub.CoR, Z.sub.CoR of the baseline PDL of the at least one tooth **62** based on the calibrated compliance matrix .sub.CRP.

[0199] In some embodiments, generating the ROM **300** further includes determining the reference mapping matrix Γ based on the plurality of center of resistance parameters X.sub.CoR, V.sub.CoR, Z.sub.CoR of the baseline PDL.

[0200] In some embodiments, generating the ROM **300** further includes determining the characteristic compliance matrix C.sub.COR of the baseline PDL of the at least one tooth **62** based on the calibrated compliance matrix C.sub.RP and the reference mapping matrix Γ . In some embodiments, the characteristic compliance matrix C.sub.COR represents the correlation between the orthodontic force parameter **430** applied at the CoR **405** with the one or more displacements at the CoR **405**.

[0201] In some embodiments, the method **800** includes removing the PDL model **410** from the PMM **400** and applying the ROM **300** at the CoR **405** to generate the updated PMM **500** of the at least one tooth **62**.

[0202] In some embodiments, the method **800** includes determining the plurality of numerical values representing compliance of the baseline PDL based on the characteristic compliance matrix C.sub.COR. In some other embodiments, the method **800** includes determining the stiffness matrix as the inverse of the characteristic compliance matrix C.sub.COR. In such embodiments, the method **800** includes determining the plurality of numerical values representing stiffness of the baseline PDL based on the stiffness matrix.

[0203] In some embodiments, the method **800** includes determining the simplified mathematical representation of the baseline PDL of the at least one tooth **62** based on the plurality of numerical values. In some embodiments, the method **800** includes determining the ROM **300** based on the simplified mathematical representation of the baseline PDL of the at least one tooth **62**.

[0204] In some embodiments, the method **800** includes scaling the ROM **300** of the PDL **64** of the at least one tooth **62**.

[0205] In some embodiments, scaling the ROM **300** includes determining the height **414** of the PDL **64** of the at least one tooth **62** based on the digital 3D representation **100**. The height **414** is different from the baseline height **411** of the baseline PDL of the similar tooth.

[0206] In some embodiments, scaling the ROM **300** includes determining the thickness **416** of the PDL **64** of the at least one tooth **62** based on the digital 3D representation **100**. The thickness **416** is different from the baseline thickness **412** of the baseline PDL of the similar tooth.

[0207] In some embodiments, scaling the ROM **300** includes determining the height **414** and the thickness **416** of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. The height **414** and the thickness **416** are different from the baseline height **411** and the baseline thickness **412**, respectively, of the baseline PDL of the similar tooth.

[0208] In some embodiments, scaling the ROM **300** includes determining the at least one corrected center of resistance parameter different from the corresponding at least one center of resistance parameter from the plurality of center of resistance parameters of the baseline PDL.

[0209] In some embodiments, the method **800** includes determining the centroid **406** of the PDL **64** of the at least one tooth **62** from the digital 3D representation **100**. The centroid **406** corresponds to the at least one corrected center of resistance parameter.

[0210] In some embodiments, the method **800** includes creating the 2D image **101** of the PDL **64** of the at least one tooth **62** from the digital 3D representation **100**. In such embodiments, the method **800** includes determining the plurality of points **105** on the PDL **64** of the at least one tooth **62** from the 2D image **101**. Further, in some embodiments, the method **800** includes connecting the plurality of points **105** to construct the triangle **106** or the semi-ellipse **107** and determining the area centroid **407** of the triangle **106** or the semi-ellipse **107**. The area centroid **407** corresponds to the at least one corrected center of resistance parameter.

[0211] In some embodiments, scaling the ROM **300** includes determining the corrected reference mapping matrix Γ_s , Γ_t , Γ_{st} based on the at least one corrected center of resistance parameter in order to map the one or more displacements at the center of resistance **405** of the at least one tooth **62** to the other locations of the at least one tooth **62**.

[0212] In some embodiments, the method **800** further includes determining the first scale factor $f_{sub.A}$ of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. The first scale factor $f_{sub.A}$ is the combination of the height **414** of the PDL **64** and the baseline height **411** of the baseline PDL. The combination of the first scale factor $f_{sub.A}$ and the characteristic compliance matrix $C_{sub.COR}$ of the baseline PDL associates the corrected characteristic compliance matrix $C_{sub.CORH}$ of the PDL **64** with the height **414** of the PDL **64**.

[0213] In some embodiments, the method **800** further includes determining the second scale factor $f_{sub.B}$ of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. The second scale factor $f_{sub.B}$ is the combination of the thickness **416** of the PDL **64** and the baseline thickness **412** of the baseline PDL. The combination of the second scale factor $f_{sub.B}$ and the characteristic compliance matrix $C_{sub.COR}$ of the baseline PDL associates the corrected characteristic compliance matrix $C_{sub.CORT}$ of the PDL **64** with the thickness **416** of the PDL **64**.

[0214] In some embodiments, the method **800** further includes determining the combined scale factor of the PDL **64** of the at least one tooth **62** in the digital 3D representation **100**. The combined scale factor $f_{sub.AB}$ is the combination of the first scale factor $f_{sub.A}$ and the second scale factor $f_{sub.B}$. The combination of the combined scale factor $f_{sub.AB}$ and the characteristic compliance matrix of the baseline PDL $C_{sub.COR}$ correlates the corrected characteristic matrix $C_{sub.CORHT}$ of the PDL **64** with the height **414** and the thickness **416** of the PDL **64**.

[0215] At step **806**, the method **800** includes generating the design of the orthodontic appliance **700** based at least partially on the ROM **300** of the PDL **64** of the at least one tooth **62**.

[0216] In some embodiments, the method **800** further includes receiving the target arrangement of the at least one tooth **62**.

[0217] In some embodiments, the method **800** further includes determining the one or more force

parameters based on the updated PMM **500** to achieve the target arrangement from the initial arrangement of the at least one tooth **62**.

[0218] In some embodiments, the method **800** further includes generating the design of the orthodontic appliance **700**, such that the orthodontic appliance **700** applies the one or more force parameters to the at least one tooth **62** to achieve the target arrangement.

[0219] In some embodiments, the method **800** further includes fabricating the orthodontic appliance **700** based on the design. In some embodiments, the orthodontic appliance **700** is fabricated using at least one of the additive manufacturing process and the subtractive manufacturing process.

[0220] In some embodiments, the method **800** further includes receiving the orthodontic appliance data **702**. The orthodontic appliance data **702** includes at least one of the orthodontic appliance material properties and the orthodontic appliance geometry.

[0221] In some embodiments, the method **800** further includes forming the virtual orthodontic appliance design **750** based on the orthodontic appliance data **702**.

[0222] In some embodiments, the method **800** further includes modifying the updated PMM **500** of the at least one tooth **62** by adding the virtual orthodontic appliance design **750** to the updated PMM **500**.

[0223] In some embodiments, the method **800** further includes simulating the interaction between the at least one tooth **62** in the updated PMM **500** and the virtual orthodontic appliance design **750**. and determining the resulting tooth arrangement. In some embodiments, the method **800** further includes comparing the resulting tooth arrangement from the simulated interaction with the target arrangement.

[0224] In some embodiments, the method **800** further includes verifying that the virtual orthodontic appliance design **750** is applying the one or more force parameters to achieve the target arrangement of the at least one tooth **62** based on the comparison and modifying the virtual orthodontic appliance design **750** if the virtual orthodontic appliance design **750** is not applying the one or more force parameters to achieve the target arrangement.

[0225] In some embodiments, the method **800** further includes displaying the virtual orthodontic appliance design **750** and modifying the virtual orthodontic appliance design **750**. In some embodiments, the method **800** further includes fabricating the orthodontic appliance **700** based on the virtual orthodontic appliance design **750**.

[0226] It will be appreciated that the arrangements presented herein may be varied in any number of aspects while still remaining within the scope of the disclosures herein.

[0227] Unless otherwise indicated, all numbers expressing feature sizes, amounts, and physical properties used in the specification and claims are to be understood as being modified by the term “about”. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the foregoing specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein.

[0228] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

Claims

1. A method for generating a design of an orthodontic appliance, the method comprising: receiving a digital three-dimensional (3D) representation of at least one tooth; accessing a Reduced Order

Model (ROM) of a periodontal ligament (PDL) of the at least one tooth in the digital 3D representation, wherein the ROM represents a mechanical response of the PDL of the at least one tooth; and generating the design of the orthodontic appliance based at least partially on the ROM of the PDL of the at least one tooth.

2. The method of claim 1, wherein the ROM comprises a simplified mathematical representation comprising a plurality of numerical values representing compliance or stiffness of the PDL of the at least one tooth with respect to three translational degrees of freedom and three rotational degrees of freedom.

3. The method of claim 1, wherein accessing the ROM comprises generating the ROM, wherein generating the ROM comprises: characterizing a mechanical response of a baseline PDL of the at least one tooth to a plurality of orthodontic force parameters; and determining one or more ROM parameters of the ROM of the baseline PDL based at least on the digital 3D representation.

4. The method of claim 3, further comprising generating a physics-based mathematical model (PMM) of the at least one tooth based on the digital 3D representation, wherein the PMM comprises at least a PDL model, and wherein the ROM is determined further based on the PMM.

5. (canceled)

6. The method of claim 4, wherein generating the ROM further comprises: determining, for the at least one tooth, a plurality of tooth movement parameters corresponding to the plurality of orthodontic force parameters applied at a reference point disposed on the at least one tooth based on the PMM; determining a calibrated compliance matrix of the baseline PDL of the at least one tooth based at least on the plurality of tooth movement parameters and the plurality of orthodontic force parameters; determining a plurality of center of resistance parameters of the baseline PDL of the at least one tooth based on the calibrated compliance matrix; determining a reference mapping matrix based on the plurality of center of resistance parameters of the baseline PDL; and determining a characteristic compliance matrix of the baseline PDL of the at least one tooth based on the calibrated compliance matrix and the reference mapping matrix.

7. The method of claim 6, further comprising: determining a plurality of numerical values representing compliance of the baseline PDL based on the characteristic compliance matrix; determining a simplified mathematical representation of the baseline PDL of the at least one tooth based on the plurality of numerical values; and determining the ROM based on the simplified mathematical representation of the baseline PDL of the at least one tooth.

8. The method of claim 6, further comprising: determining a stiffness matrix as an inverse of the characteristic compliance matrix; determining a plurality of numerical values representing stiffness of the baseline PDL based on the stiffness matrix; determining a simplified mathematical representation of the baseline PDL of the at least one tooth based on the plurality of numerical values; and determining the ROM based on the simplified mathematical representation of the baseline PDL of the at least one tooth.

9. The method of claim 6, wherein each of the plurality of tooth movement parameters comprises a plurality of tooth displacements at the reference point due to an application of a corresponding orthodontic force parameter from the plurality of orthodontic force parameters at the reference point.

10. The method of claim 6, wherein the calibrated compliance matrix provides a correlation between the plurality of orthodontic force parameters and the plurality of tooth movement parameters, and wherein the characteristic compliance matrix represents a correlation between an orthodontic force parameter applied at a center of resistance with one or more displacements at the center of resistance.

11. The method of claim 4, further comprising: removing the PDL model from the PMM; and applying the ROM at the center of resistance to generate an updated PMM of the at least one tooth. receiving a target arrangement of the at least one tooth; and determining one or more force parameters based on the updated PMM to achieve the target arrangement from an initial

arrangement of the at least one tooth, wherein the digital 3D representation represents the initial arrangement of the at least one tooth.

12. The method of claim 1, further comprising generating the design of the orthodontic appliance, such that the orthodontic appliance applies the one or more force parameters to the at least one tooth to achieve the target arrangement.

13. (canceled)

14. The method of claim 12, further comprising: receiving a target arrangement of the at least one tooth; receiving an orthodontic appliance data, wherein the orthodontic appliance data comprises at least one of orthodontic appliance material properties and orthodontic appliance geometry; and forming a virtual orthodontic appliance design based on the orthodontic appliance data.

15. The method of claim 14, further comprising modifying the updated PMM of the at least one tooth by adding the virtual orthodontic appliance design to the updated PMM and simulating an interaction between the at least one tooth in the updated PMM and the virtual orthodontic appliance design and determining a resulting tooth arrangement.

16. The method of claim 15, further comprising: verifying that the virtual orthodontic appliance design is applying one or more force parameters to achieve the target arrangement of the at least one tooth based on the comparison; and modifying the virtual orthodontic appliance design if the virtual orthodontic appliance design is not applying the one or more force parameters to achieve the target arrangement, fabricating the orthodontic appliance based on the virtual orthodontic appliance design.

17. The method of claim 6, further comprising scaling the ROM of the PDL of the at least one tooth, wherein scaling the ROM comprises: determining a height of the PDL of the at least one tooth based on the digital 3D representation, wherein the height is different from a baseline height of the baseline PDL of a similar tooth; determining at least one corrected center of resistance parameter different from corresponding at least one center of resistance parameter from the plurality of center of resistance parameters of the baseline PDL; and determining a corrected reference mapping matrix based on the at least one corrected center of resistance parameter in order to scale the ROM of the PDL of the at least one tooth from that of the baseline PDL.

18. The method of claim 17, further comprising determining a first scale factor of the PDL of the at least one tooth in the digital 3D representation, wherein the first scale factor is a combination of the height of the PDL and the baseline height of the baseline PDL, and wherein a combination of the first scale factor and the characteristic compliance matrix of the baseline PDL associates a corrected characteristic compliance matrix of the PDL with the height of the PDL.

19. (canceled)

20. The method of claim 17, further comprising: creating a two-dimensional (2D) image of the PDL of the at least one tooth from the digital 3D representation; determining a plurality of points on the PDL of the at least one tooth from the 2D image; connecting the plurality of points to construct a triangle or a semi-ellipse; and determining an area centroid of the triangle or the semi-ellipse, wherein the area centroid corresponds to the at least one corrected center of resistance parameter.

21. The method of claim 6, further comprising scaling the ROM of the PDL of the at least one tooth, wherein scaling the ROM comprises: determining a thickness of the PDL of the at least one tooth based on the digital 3D representation, wherein the thickness is different from a baseline thickness of the baseline PDL of a similar tooth; determining at least one corrected center of resistance parameter different from corresponding at least one center of resistance parameter from the plurality of center of resistance parameters of the baseline PDL; and determining a corrected reference mapping matrix based on the at least one corrected center of resistance parameter in order to map the one or more displacements at the center of resistance of the at least one tooth to the other locations of the at least one tooth.

22. (canceled)

23. The method of claim 6, further comprising scaling the ROM of the PDL of the at least one tooth, wherein scaling the ROM comprises: determining a height and a thickness of the PDL of the at least one tooth in the digital 3D representation, wherein the height and the thickness are different from a baseline height and a baseline thickness of the baseline PDL of a similar tooth; determining at least one corrected center of resistance parameter different from corresponding at least one center of resistance parameter from the plurality of center of resistance parameters of the baseline PDL; and determining a corrected reference mapping matrix based on the at least one corrected center of resistance parameter in order to map the one or more displacements at the center of resistance of the at least one tooth to the other locations of the at least one tooth.

24. The method of claim 6, further comprising determining a combined scale factor of the PDL of the at least one tooth in the digital 3D representation, wherein the combined scale factor is a combination of the first scale factor and the second scale factor, and wherein a combination of the combined scale factor and the characteristic compliance matrix of the baseline PDL associates a corrected characteristic matrix of the PDL with the height and the thickness of the PDL.
